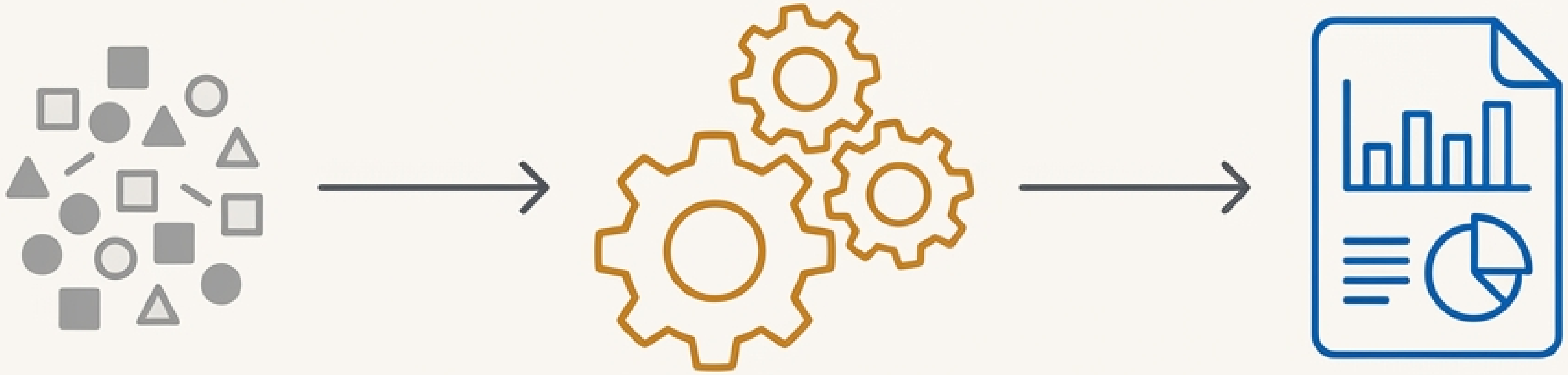
Automating EDA with ydata-profiling

A Beginner's Guide to Generating Instant Data Insights in Python



The Challenge of Manual Data Exploration

- Analyzing new datasets can be slow and repetitive.
- Imagine a dataset with 10,000 rows and 50 columns. Where do you start?
- Writing code for each analysis step is time-consuming:
  - Checking data types and unique values.
  - Finding missing data points.
  - Calculating correlations.
  - Visualizing distributions for every variable.
- This initial exploration, or EDA, is critical but can take hours or days.

The Solution: One-Line Exploratory Data Analysis

ydata-profiling (formerly Pandas Profiling) is a Python library that automates EDA.

- Generates a complete, interactive HTML report for your dataset.
- Summarizes key statistics, data types, missing values, and more.
- Reduces the complex task of initial EDA to just a few lines of code.
- Helps you understand the structure of your data instantly.

```
import pandas as pd
df = pd.read_csv(...)
ProfileReport(df)
```

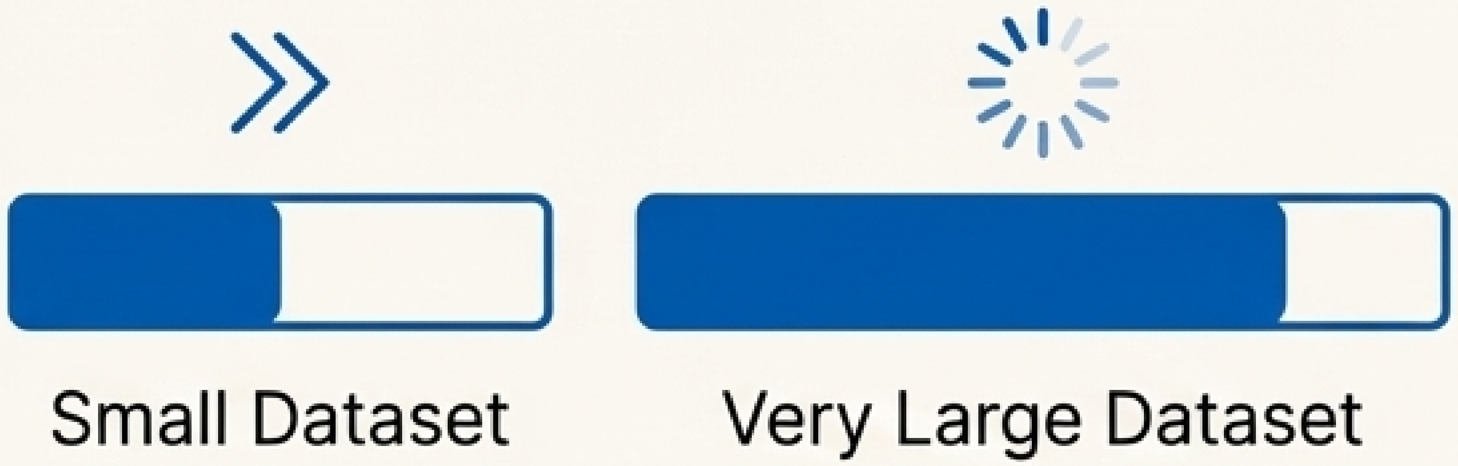
Key Advantages and A Word of Caution

Advantages

- **Ease of Use**: Generate a comprehensive report with minimal code.
- **Time-Saving**: Automates hours of manual analysis work.
- **Interactive Reports**: Provides easy-to-navigate HTML reports to dig deeper into variables.

Disadvantages

- **Performance**: Report generation can be slow for very large datasets. The library is best suited for small to medium-sized data.



Step 1: Installation

To get started, install the library using your preferred package manager.

```
# Using Pip
pip install ydata-profiling
```

```
# Using Conda
conda install -c conda-forge ydata-profiling
```

**Make sure you have pandas installed as well.*

Step 2: Generating Your First Report

In just a few lines, you can import your data and create a detailed profile.

```python
# 1. Import necessary libraries
import pandas as pd
from ydata_profiling import ProfileReport

# 2. Load your data into a DataFrame
df = pd.read_csv("trending-books.csv")

# 3. Create the report object
# The 'title' is customizable
profile = ProfileReport(df, title="Trending Books Report")

# 4. Display the report in your notebook
profile.to_notebook_iframe()
```

Summarize dataset: 100%
Generate report structure: 100%
Render HTML: 100%

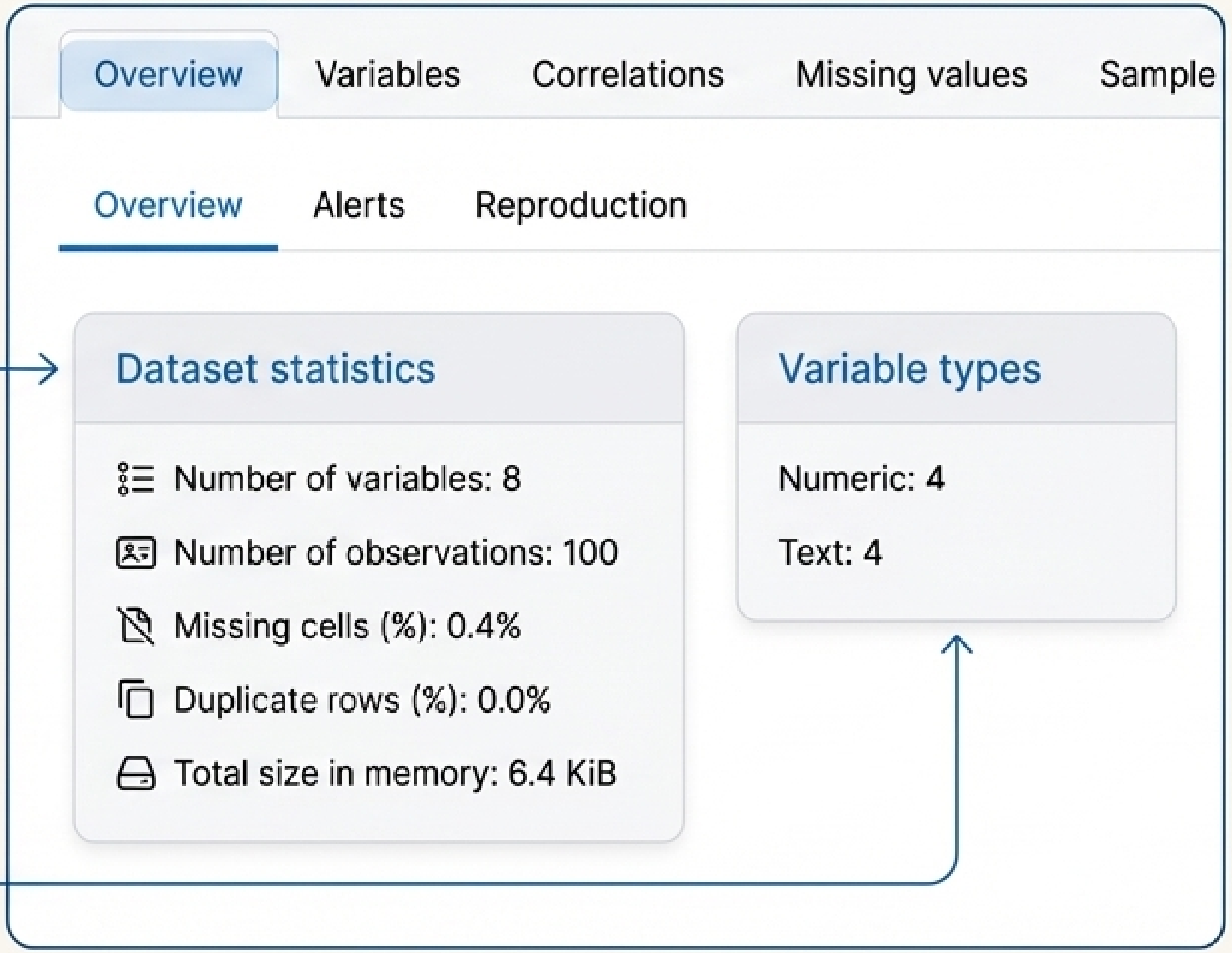
Exploring the Report: The Overview Tab

The report opens with a **high-level summary** of your **entire dataset.**

Dataset statistics: Gives you the essentials at a glance.

- Number of variables (columns)
- Number of observations (rows)
- Missing cells (%)
- Duplicate rows (%)
- Total size in memory

Variable types: Shows the breakdown of column types (e.g., Numeric, Text).



Exploring the Report: Alerts & Reproduction

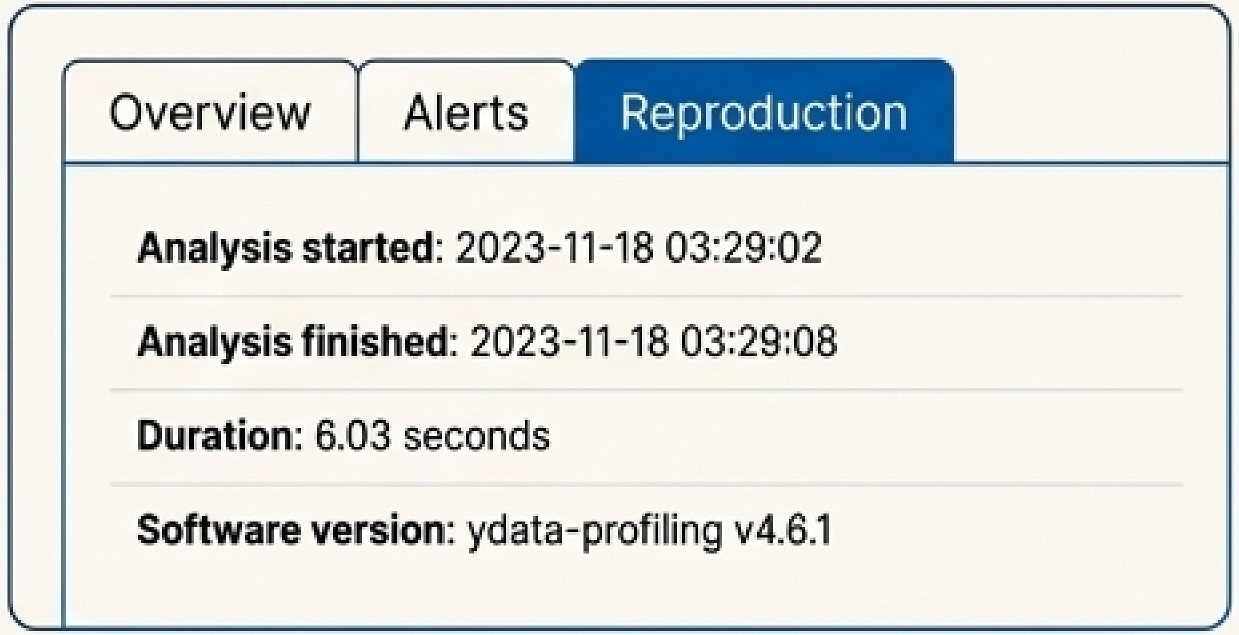
The Overview also provides important warnings and analysis metadata.

Alerts

- Highlights potential data quality issues automatically.
- Examples: high correlation, missing values, uniform distributions, or unique values.

Reproduction

- Provides technical details about the analysis.
- Includes analysis duration, start/end times, and the `ydata-profiling` version used.

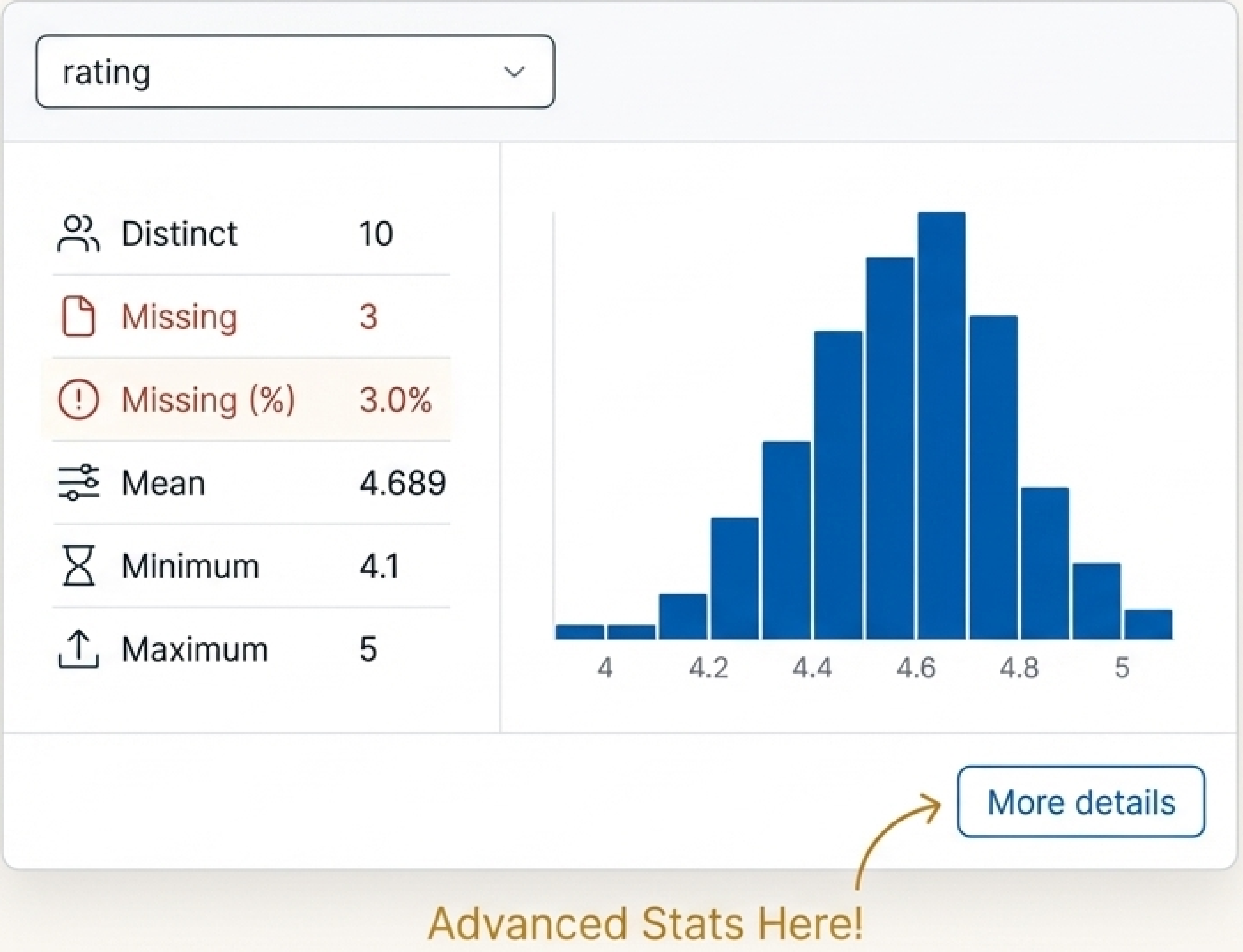


Exploring the Report: Detailed Variable Analysis

Select any column from the dropdown to get a detailed breakdown.

Key Information for each Variable

- **Statistics**: Distinct values, missing values, mean, min, max.
- **Distribution**: A histogram provides a quick visual understanding of the data's shape.
- **More details**: A button reveals advanced statistics like median, standard deviation, and more.



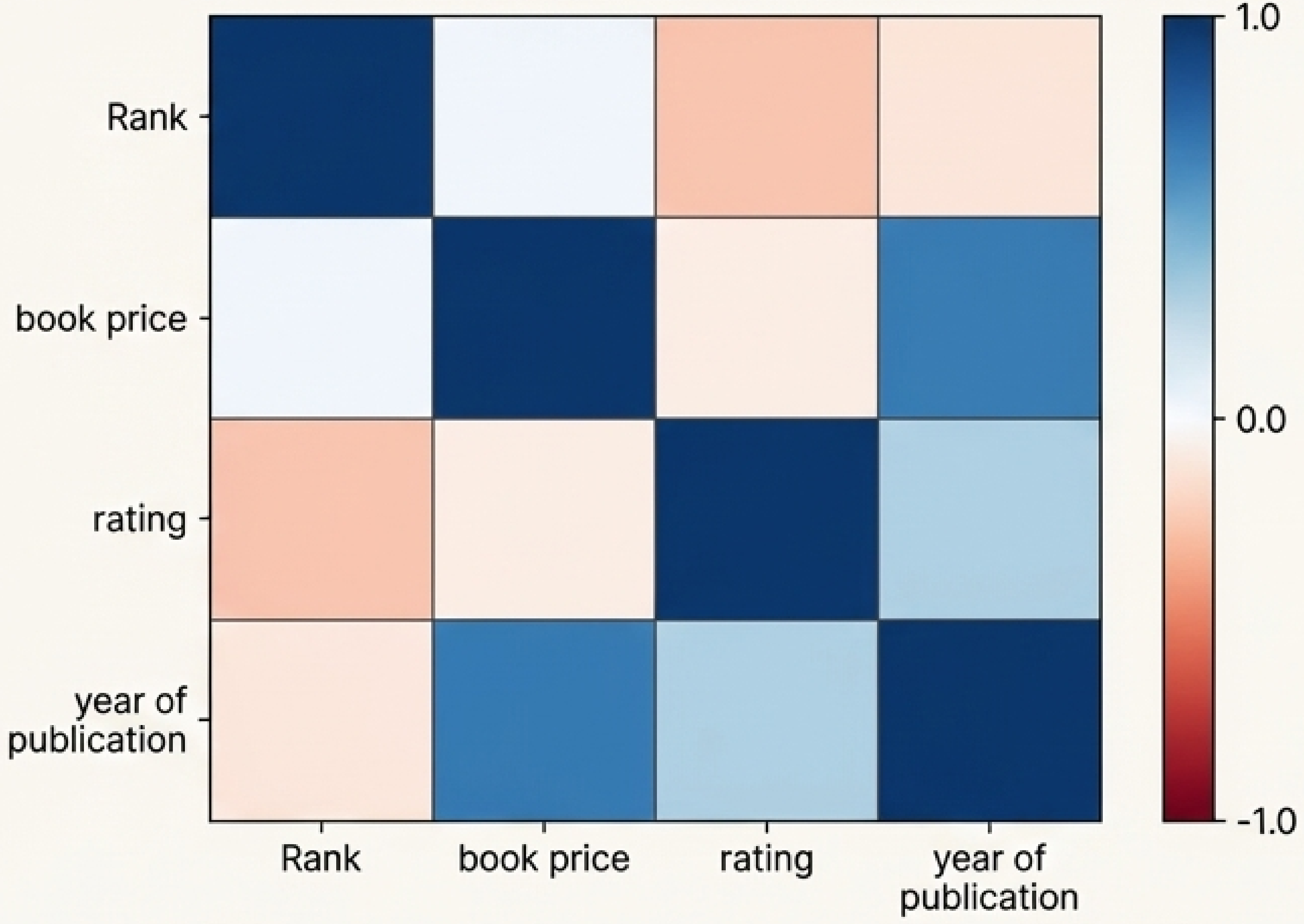
Exploring the Report: Understanding Correlations

The 'Correlations' tab visualizes the relationship between numeric variables using a heatmap.

How to Read the Heatmap

- Strong positive correlation. As one variable increases, the other tends to increase.
- Strong negative correlation. As one variable increases, the other tends to decrease.

Weak or no correlation.

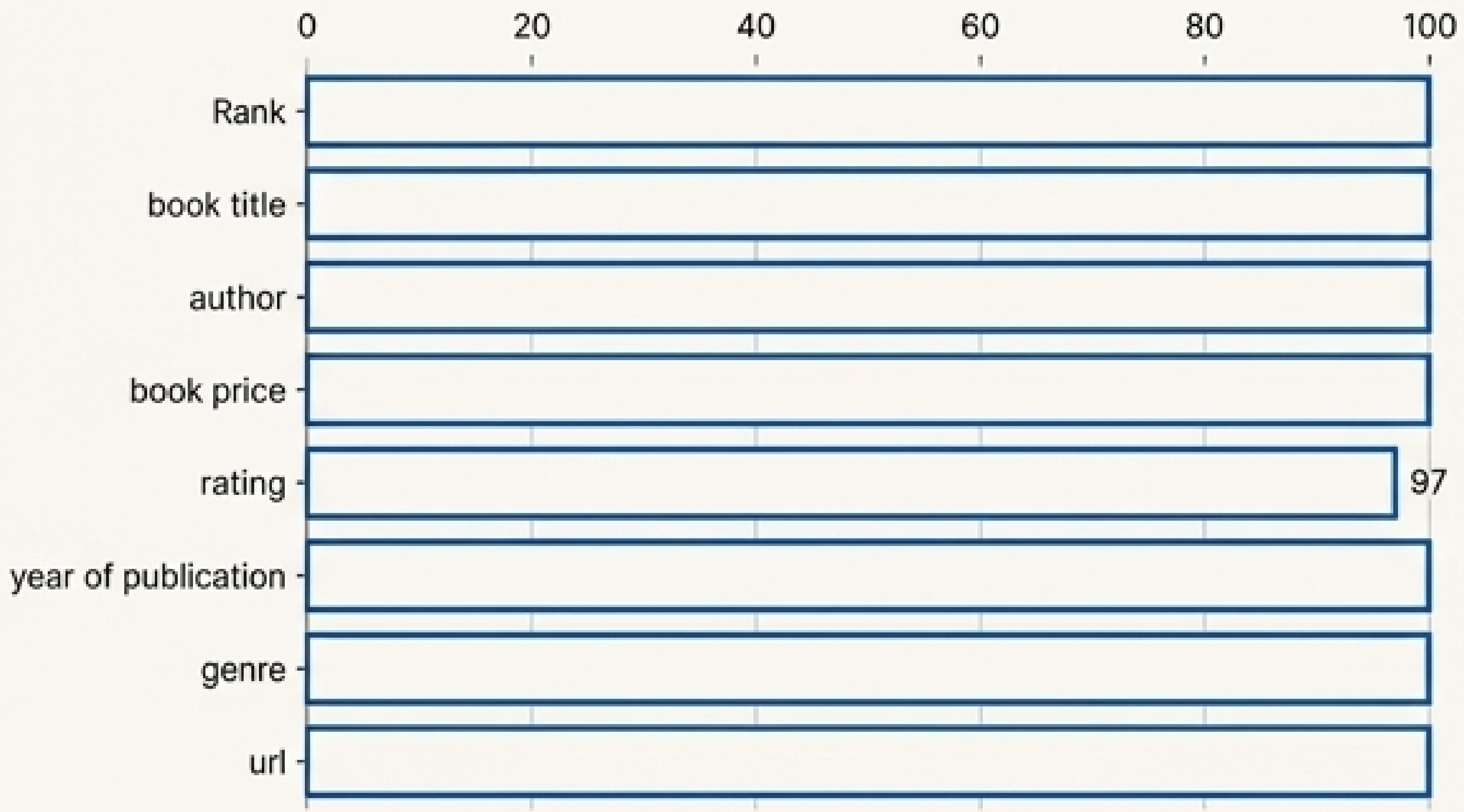


Exploring the Report: Visualizing Missing Values

The “Missing values” section provides two clear ways to see where data is missing.

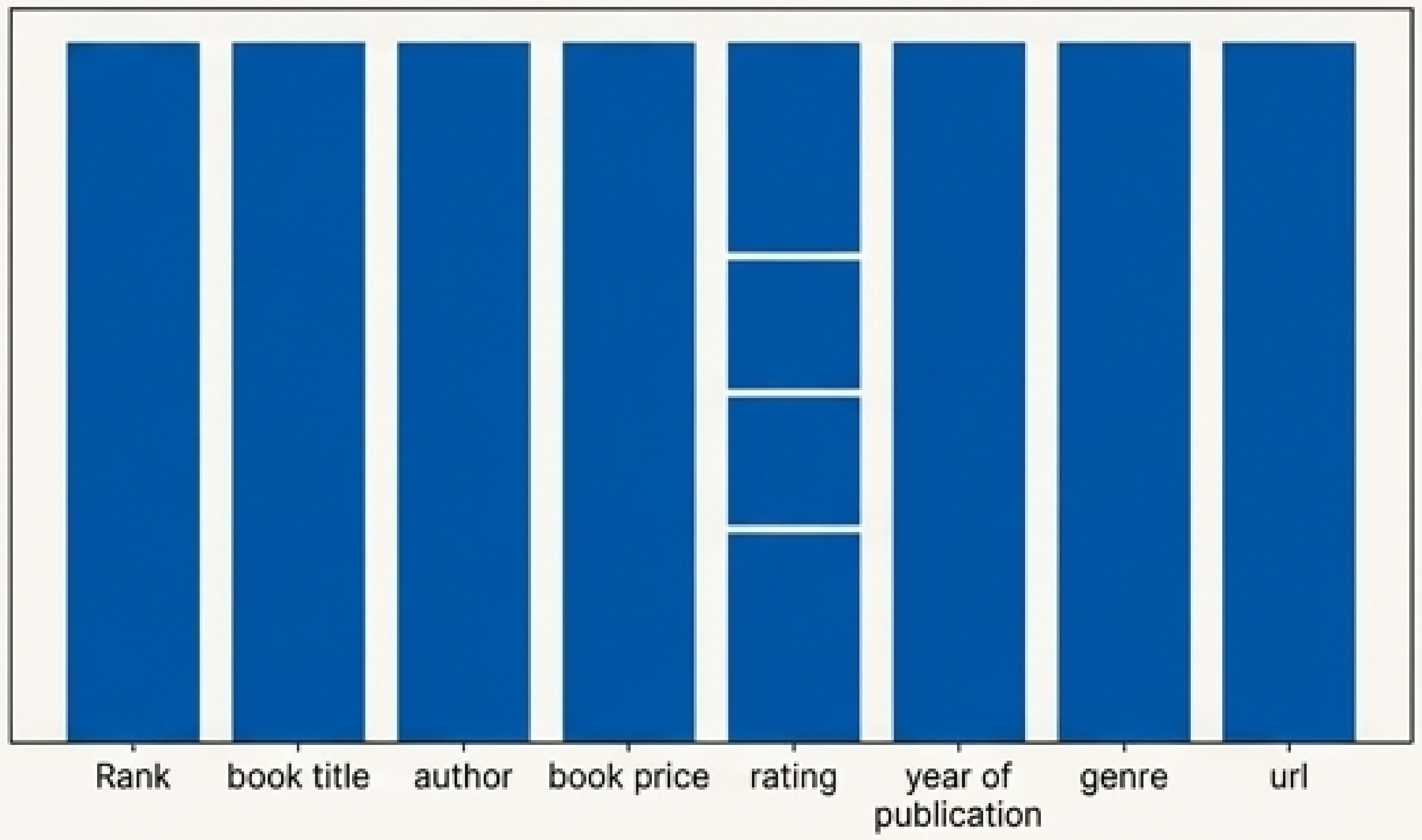
Count Plot

- A bar chart showing the count of non-missing values for each column.
- Quickly identifies columns with missing data.



Matrix

- A matrix plot showing the location of missing values across all rows.
- White lines indicate missing data points.



Exploring the Report: The Sample Tab

The 'Sample' tab allows you to view the raw data directly within the report.

- Shows the first 10 rows of the dataset.
- Shows the last 10 rows of the dataset.
- Useful for a quick sanity check of the data's content and format.

First rows | Last rows

Rank	book title	book price	rating	author	year of publication
1	Iron Flame (The Empyrean, 2)	18.42	4.1	Rebecca Yarros	2023
2	The Woman in Me	20.93	4.5	Britney Spears	2023
3	My Name is Barbra	31.50	4.5	Barbra Streisand	2023
4	The Ballad of Songbirds and Snakes (A Hunger Games Novel)	14.99	4.2	Suzanne Collins	2020
5	Fourth Wing (The Empyrean, 1)	16.88	4.3	Rebecca Yarros	2023

Step 3: Saving and Sharing Your Report

You can save the interactive report as a standalone HTML or JSON file using the `to_file()` method.

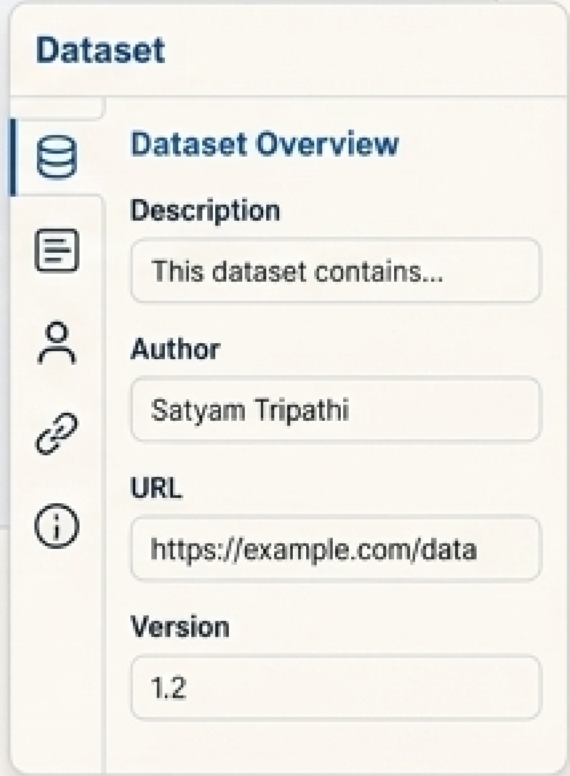
```
# Save the report to an HTML file
profile.to_file("trending_books_report.html")

# Or save the report data to a JSON file
profile.to_file("trending_books_data.json")
```

Advanced Uses: Customizing Your Report

Add metadata for context or use minimal mode for large datasets.

Add author, a description, and other details to the report's overview.

```
profile = ProfileReport(df,
  dataset={
    "description": "...",
    "author": "Satyam Tripathi",
    ...
})
```



For faster performance, generate a minimal report that skips expensive computations.

```
profile = ProfileReport(df, minimal=True)
```

NotebookLM

Summary: Your New EDA Workflow

- **Problem:** Manual EDA is slow and repetitive. 
- **Solution:** `ydata-profiling` automates data exploration with one command. 
- **Process:**
  1. Install the library.
  2. Load data into a pandas DataFrame.
  3. Generate the `ProfileReport`.
  4. Explore the interactive HTML report to understand variables, correlations, and missing data.
  5. Save and share your findings.
- **Key Benefit: Go from raw data to deep insights in minutes, not hours.**

Knowledge Check

Question 1: What is the primary disadvantage of using `ydata-profiling`?

A) It is difficult to install.

B) It can be slow on very large datasets.

C) The reports are not interactive.

Question 2: Which method is used to display the report inside a Jupyter Notebook?

A) `.show_report()`

B) `.to_notebook_iframe()`

C) `.render_notebook()`

Question 3: In the report, which section provides a heatmap to show relationships between numeric variables?

A) Overview

B) Variables

C) Correlations